A σ-Essential Contextual State in ZFC: Localization and the Product Ulam Witness

Patrick S. Mahon*

Abstract

A two-valued state on a concrete quantum logic of subsets of a set Ω assigns a definite truth value to each event, coherently across overlapping measurement contexts. On a Boolean σ-algebra every finitely coherent local pattern of such assignments extends to a global σ-additive two-valued state. We ask whether the non-distributive case can differ: does there exist a concrete σ-complete non-Boolean orthomodular poset carrying a finitely coherent local pattern that extends to *no* global σ-additive two-valued state — a *σ-essential contextual state*? We first localize the question: a witness exists iff the pattern is finitely coherent while no point evaluation and no non-principal σ-additive state extends it, and we map its boundary — empty in the Polish-representable case (Derr–Williamson), irreducible to the Hilbert-space dichotomy of Blecher–Weaver, irreducible to a measurable cardinal. We then answer it affirmatively in ZFC. The witness is a *product Ulam carrier* on $\Omega = \omega_1 \times \{1, 2, 3, 4\}$: an Ulam matrix imported as generators makes every σ-additive two-valued state a point evaluation, a Specker triple of fibre events has empty kernel, and a trace-invariant normal form shows the two mechanisms cannot interact, so that a finitely additive extension survives, built from an arbitrary ultrafilter on $\mathcal{P}(\omega_1)/\text{ctble}$ through a parity code. Consequently the target sentence carries no large-cardinal strength. The quotient of the carrier by the countable ideal is a σ-complete non-Boolean orthomodular poset with finitely additive two-valued states but no σ-additive ones at all. The lattice (OML) form of the question remains open, and we conjecture it fails: latticehood appears to force σ-liftability.

1 Introduction

A finite system of measurement contexts may admit a consistent assignment of truth values to events that no single underlying state realises, the algebraic signature of quantum contextuality. When the contexts are countable and one demands countable additivity, a sharper question arises: can a *finitely* coherent pattern fail to extend to any global σ-additive state, with the failure visible only at the σ-level? This paper isolates that phenomenon for concrete quantum logics, reduces it to an irreducible set-theoretic core, locates that core against the measure-theoretic and operator-algebraic literature — and then resolves it affirmatively, in ZFC, by an explicit construction: the *product Ulam carrier*. We work throughout with two-valued states, the non-distributive analogue of a $\{0, 1\}$-valued measure.

A *concrete σ-complete orthomodular poset* (concrete σ-orthoposet, σ-class) is a family $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ with

$$\varnothing, \Omega \in \mathcal{L}; \qquad A \in \mathcal{L} \implies A^{\perp} := \Omega \setminus A \in \mathcal{L}; \qquad \{A_n\} \subseteq \mathcal{L} \text{ pairwise disjoint} \implies \bigcup_n A_n \in \mathcal{L},$$

ordered by inclusion. This is the σ-class of Navara–Pták [19], a Gudder concrete quantum logic [11, 20]: the maximal Boolean subalgebras ("blocks") overlap but their union need not be Boolean. (An earlier version of this paper wrote "lattice" where "poset" is meant; the σ-class

*Research Computing Group, Simon Fraser University, Burnaby, BC, Canada. pmahon@sfu.ca

axioms above, which are what every argument uses, do not provide lattice meets, and the witness constructed in §5 is provably *not* a lattice. The lattice form of the problem remains open; see §6.) A *two-valued state* is $s\colon \mathcal{L} \to \{0,1\}$ with $s(\Omega) = 1$ that is additive over orthogonal joins lying in its domain: whenever $\{A_n\} \subseteq \mathcal{L}$ is a *finite* pairwise disjoint family with $\bigcup_n A_n \in \mathcal{L}$,

$$s(\textstyle\bigcup_n A_n) = \sum_n s(A_n)$$

(in particular $s(A) + s(A^\perp) = 1$ on complement pairs); it is *σ-additive* when this holds for countable such families. The same additivity is required of states on a sub-orthoposet $B \subseteq \mathcal{L}$, over orthogonal joins that exist *in* B. The *point evaluations* $\delta_\omega(A) := \mathbf{1}[\omega \in A]$ are σ-additive two-valued states; call these *Dirac* and all others *non-Dirac*. For a finite $\perp$-closed sub-orthoposet $B \subseteq \mathcal{L}$ and a two-valued state s_0 on B, s_0 *extends* to s on $\mathcal{L}$ if $s \restriction B = s_0$.

Definition 1.1 (finite coherence)**.** Let $B \subseteq \mathcal{L}$ be a finite $\perp$-closed sub-orthoposet and s_0 a two-valued state on B. Say s_0 is *finitely coherent on* $\mathcal{L}$ if s_0 extends to a (finitely additive) two-valued state on all of $\mathcal{L}$.

Definition 1.2 (σ-essential contextual state)**.** A *σ-essential contextual state* on $\mathcal{L}$ is a two-valued state s_0 on a finite $\perp$-closed sub-orthoposet $B \subseteq \mathcal{L}$ that is finitely coherent on $\mathcal{L}$ but extends to *no* global σ-additive two-valued state on $\mathcal{L}$. A concrete σ-orthoposet $\mathcal{L}$ *carries a σ-essential contextual state* (is a *witness*) if such an s_0 exists. We require $\mathcal{L}$ to be non-Boolean and *essentially irreducible*: the quotient of $\mathcal{L}$ by the σ-ideal of countable sets has centre $\{0,1\}$; see Remarks 1.5 and 1.6.

Remark 1.3 (Why the coherence clause is necessary)**.** Without Definition 1.1 the definition of a witness is satisfiable on a Boolean carrier, trivially and against its intent. Let $\Omega_7 = \{\pm\}^3 \setminus \{(+,+,+)\}$, let $\mathcal{L} = \mathcal{P}(\Omega_7)$, and let B be the $\perp$-closure of the three coordinate events $A_i = \{x : x_i = +\}$. All twelve pairwise intersections among $\{A_i, A_i{}^\perp\}$ $(i \neq j)$ are nonempty, so the only additivity constraints inside B are the complement pairs, and $s_0(A_1) = s_0(A_2) = s_0(A_3) = 1$ is a two-valued state on B. But $A_1 \cap A_2 \cap A_3 = \varnothing$, and on a finite power set every two-valued state is Dirac, so s_0 extends to no σ-additive state — on a Boolean σ-algebra. The pattern is Specker's parable (pairwise but not jointly) embedded as a sub-orthoposet, and locality of coherence cannot distinguish distributive from non-distributive carriers. The phenomenon of interest is a failure *at σ*: finite additivity achievable globally, countable additivity not. Definition 1.1 makes this the definition, and Proposition 2.1 shows the Boolean baseline is then restored.

The phenomenon is a failure of σ-level compactness. Wright's theorem [23], that every two-valued state on a *finite* OML is finitely realised, forbids any *finite* witness, so the question is whether a coherent finite pattern can fail to globalise once σ-additivity is imposed. On a Boolean carrier it cannot (§2). The main result of this paper is that on a non-distributive carrier it can, without hypotheses:

Theorem 1.4 (Main; proved as Theorem 5.18)**.** *In ZFC there exist a set Ω with $|\Omega| = \aleph_1$, a concrete σ-complete orthomodular poset $\mathcal{L} \subseteq \mathcal{P}(\Omega)$, a finite $\perp$-closed $B \subseteq \mathcal{L}$, and a two-valued state s_0 on B such that s_0 extends to a finitely additive two-valued state on $\mathcal{L}$ but to no σ-additive one. Moreover every σ-additive two-valued state on $\mathcal{L}$ is Dirac, and $\mathcal{L}$ is non-Boolean, essentially irreducible, non-segregated, and not Polish-representable.*

Results. §2: the Boolean baseline (Prop. 2.1, now three obstructions). §3: the *Localization Theorem* (Thm. 3.2): a witness exists iff (0) s_0 is finitely coherent while (i) no Dirac and (ii) no non-Dirac σ-additive two-valued state extends it, with (i) freely arrangeable, localising all weight to the interaction of (0) with (ii); stripped of lattice scaffolding this is a selection problem (Q. 3.6). §4: the boundary map — empty above the Polish case, irreducible to Blecher–Weaver, irreducible to a measurable cardinal, with the segregation obstruction repaired and proved (Lemma 4.3). §5: the witness. The carrier is a product $\omega_1 \times \{1, 2, 3, 4\}$; an Ulam matrix imported as generators makes every σ-additive two-valued state Dirac (Theorem 5.13); a Specker triple of fibre events has empty kernel, so no Dirac extends it; and a trace-invariant normal form (Theorem 5.8) shows the two mechanisms cannot interact, so that a finitely additive extension can be built from an arbitrary ultrafilter on $\mathcal{P}(\omega_1)/\text{ctble}$ through a parity code (Theorem 5.16). §6: consequences — Ψ is a ZFC theorem carrying no large-cardinal strength, the boundary map read as an explanation of the witness's shape, a quotient carrier with *no* σ-additive two-valued states at all, and the problems that remain open, foremost the lattice (OML) form.

Remark 1.5 (Irreducibility is not vacuous)**.** The carrier class is inhabited: the horizontal sum MO_κ (κ orthogonal atom-pairs with $0, 1$) is concrete, non-Boolean, and irreducible (centre $\{\varnothing, \Omega\}$) for $\kappa \geq 2$ [14], and σ-complete for $\kappa = \omega$. Concreteness does not force a non-trivial centre: by the Zierler–Schlessinger/Gudder criterion it is exactly order-determination by two-valued states [11, 12], unrelated to reducibility. (MO_κ itself is excluded as a witness by Lemma 4.3.)

Remark 1.6 (Why irreducibility is required modulo the countable ideal)**.** Literal irreducibility (centre $\{\varnothing, \Omega\}$) is unavailable to any carrier containing all singletons: a countable set is compatible with every element of a σ-class containing singletons (Lemma 5.3), so the countable and co-countable sets are always central. The only rigidity mechanism known to us — and the one the witness uses — requires singletons (Theorem 5.13), so the amendment is forced by the method, not chosen for convenience. Essential irreducibility retains the intent: the contextuality may not be carried by a central factor surviving in the quotient, and for the witness the quotient centre is computed exactly (Corollary 5.12).

Verification status. The construction of §5 was developed jointly with a large language model (see Acknowledgements), hand-checked by two independent adversarial passes, and subsequently *formalized in Lean 4 in full*: the main theorem is machine-checked end-to-end in the author's repository (`formalization/QuerySystem/`, modules `SigmaEssentialAmended` and `UlamWitness*`; final statement `psiAmended_ZFC`), with zero `sorry`s and axioms exactly `propext`, `Classical.choice`, `Quot.sound` — i.e. ZFC, no additional axioms and no large cardinals. The formalization covers the Ulam matrix (Lemma 5.1), the normal form and the full disjointness table (Theorem 5.8, Lemma 5.6 including case (IV)), rigidity (Theorem 5.13), the vote state and its additivity (Theorem 5.16), and the assembly at ω_1 (Theorem 5.18); the Ω_7 example of Remark 1.3 is also machine-checked, so the necessity of the coherence clause is itself certified. The remaining trust surface is definitional fidelity (that the formal definitions transcribe Definitions 1.1–1.2), a finite reading exercise.

2 The Boolean baseline

On a Boolean σ-algebra, Definition 1.2 is never satisfied. The proof isolates the obstructions a witness must overcome at once; this is the baseline the non-distributive case must escape.

Proposition 2.1 (No witness in the Boolean case)**.** *Let $\mathcal{L} \subseteq \mathcal{P}(\Omega)$ be a Boolean σ-algebra and $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet with a two-valued state s_0 that is finitely coherent on $\mathcal{L}$. Then s_0 extends to a Dirac, hence to a global σ-additive two-valued state on $\mathcal{L}$.*

Proof. Let $\mu \supseteq s_0$ be a global finitely additive two-valued state on $\mathcal{L}$ (Definition 1.1) and $B_0 \subseteq \mathcal{L}$ the finite Boolean subalgebra generated by B. The atoms of B_0 partition Ω; finite additivity of μ over this finite partition yields a unique atom $a \in B_0$ with $\mu(a) = 1$, and a, being a nonzero element of the concrete algebra B_0, is a nonempty subset of Ω. Every element of B_0 either contains a or is disjoint from it, so for any $\omega \in a$, $\delta_\omega \restriction B_0 = \mu \restriction B_0$, whence

$$K(s_0) := \bigcap\{A \in B : s_0(A) = 1\} \;\supseteq\; a \;\neq\; \varnothing$$

(we call $K(s_0)$ the *kernel* of s_0), $\delta_\omega \restriction B = s_0$ by Lemma 3.1, and δ_ω is σ-additive. □

Note the division of labour: coherence supplies the global finitely additive μ, and *distributivity converts it into a point* — on a Boolean carrier the atom carrying μ is a common point of the s_0-true events. Three obstructions combine, all of which a witness must defeat:

(a) *(coherence)* the pattern must extend finitely at all (Remark 1.3: empty-kernel patterns exist even on Boolean carriers, but there they are incoherent);

(b) *(used above)* on a Boolean carrier any finitely additive global extension concentrates on a nonempty atom, a common point. A witness breaks this: the events lie in incompatible blocks with empty meet, so no Dirac extends s_0, and the finitely additive extension must be genuinely non-Dirac;

(c) *(complementary)* away from a measurable cardinal, every global σ-additive two-valued state on a Boolean σ-algebra is Dirac [21, 13]; so even where (b) fails, no *non*-Dirac σ-additive state repairs the gap on a Boolean carrier.

These pull against one another: breaking (b) requires non-distributivity, which is precisely what makes (c) escapable by admitting non-Dirac global states. A witness must break (b) while leaving no such repair available — and, by (a), must do so without breaking finite coherence. The construction of §5 is engineered around exactly this triple bind.

3 The Localization Theorem

When a Dirac extends s_0 is elementary.

Lemma 3.1. *Let $B \subseteq \mathcal{L}$ be a finite $\perp$-closed sub-orthoposet with two-valued state s_0. A point evaluation δ_ω satisfies $\delta_\omega \restriction B = s_0$ iff $\omega \in K(s_0) = \bigcap\{A \in B : s_0(A) = 1\}$. Consequently no Dirac extends s_0 iff $K(s_0) = \varnothing$.*

Proof. By definition,

$$\delta_\omega \restriction B = s_0 \iff \big(\omega \in A \iff s_0(A) = 1\big) \text{ for all } A \in B.$$

Split B by the value of s_0. The true constraints give $\omega \in A$ for every A with $s_0(A) = 1$, i.e. $\omega \in K(s_0)$. A false A ($s_0(A) = 0$) is handled by its complement: $\perp$-closure gives $A^\perp \in B$, and additivity gives $s_0(A^\perp) = 1 - s_0(A) = 1$, so $A^\perp$ is true and the false constraint reduces to a true one, already subsumed in $K(s_0)$. Hence the false constraints add nothing and $\delta_\omega \restriction B = s_0 \iff \omega \in K(s_0)$. □

Theorem 3.2 (Localization). *Let $\mathcal{L}$ be a concrete σ-orthoposet, $B \subseteq \mathcal{L}$ a finite $\perp$-closed sub-orthoposet, and s_0 a two-valued state on B. Then s_0 is a σ-essential contextual state iff*

(0) *s_0 is finitely coherent on $\mathcal{L}$;*

() *$K(s_0) = \varnothing$ (no point evaluation extends s_0); and*

(i) *no non-Dirac σ-additive two-valued state on $\mathcal{L}$ extends s_0.*

Moreover clauses (0) *and* (i) *are jointly freely arrangeable: there is a concrete σ-orthoposet and a finitely coherent s_0 with $K(s_0) = \varnothing$.*

Proof. The Dirac/non-Dirac dichotomy on global σ-additive two-valued states is exhaustive, so s_0 has no σ-additive extension iff it has no Dirac extension and no non-Dirac extension. By Lemma 3.1 the first conjunct is (i); the second is (ii); clause (0) is Definition 1.2's coherence requirement. For joint arrangeability of (0) and (i), the Navara–Pták construction (Example 3.4) realises $K(s_0) = \varnothing$ with a global extension that is even σ-additive. $\square$

Remark 3.3 (Scope: a decomposition, not a reduction)**.** The equivalence is a decomposition, not a reduction to a separate named problem: the Dirac/non-Dirac split being exhaustive, it merely unfolds the definition of witness. Its content is that (0)+(i) is freely arrangeable, so all weight of Definition 1.2 sits in the conjunction of (0) with (ii): coherence must survive while every non-Dirac σ-additive repair is destroyed. The two clauses pull in opposite directions — the structure rich enough to kill all non-Dirac σ-states threatens to kill the finitely additive extension too — and the witness of §5 succeeds by provably decoupling them (Theorem 5.8).

The Navara–Pták construction supplies clauses (0)+(i) and shows (ii) is binding.

Example 3.4 (Navara–Pták [19])**.** Let $\Omega = Q_0 \times Q_0$ with $Q_0 = \mathbb{Q} \cap (0,1)$, and let $f, g \colon \Omega \to \mathbb{R}$ be the coordinate projections. Let $\mathcal{L} = \mathcal{L}_{f,g}$ be the least σ-class containing the inverse images of Borel sets under f and g. Set $C_f = f^{-1}(\{1/3\})$, $C_g = g^{-1}(\{1/2\})$, $C_{f+g} = (f+g)^{-1}(\{1/2\})$. Define $m(A) = 1$ iff A contains one of C_f, C_g, C_{f+g}. Navara and Pták show m is a (two-valued, σ-additive) state on $\mathcal{L}$, and that since $C_f \cap C_g \cap C_{f+g} = \varnothing$ the state m is not concentrated at any point.

Remark 3.5 (The example is not itself a witness)**.** Let B be the $\perp$-closure of $\{C_f, C_g, C_{f+g}\}$ and let s_0 be the pattern assigning each of the three the value 1. Since $C_f \cap C_g \cap C_{f+g} = \varnothing$, we have $K(s_0) = \varnothing$, so clause (i) holds: no point evaluation extends s_0. Yet the Navara–Pták state satisfies $m \restriction B = s_0$ and is non-Dirac and σ-additive, so clause (ii) fails and s_0 is not σ-essential: the construction supplies its own rescuer, so clauses (0)+(i) cannot suffice alone. The repair is available because $\mathcal{L}_{f,g}$ is generated by two compatible (commuting) observables, Boolean enough to support a non-Dirac global state through the point-level gap. A witness requires the opposite: a carrier on which no non-Dirac state repairs the gap — while, per clause (0), a finitely additive one still must.

Stripped of lattice scaffolding, the core is a selection problem. Each block is a maximal compatible context, equivalently a countable partition Π_α of Ω into elements of $\mathcal{L}$; a two-valued state selects one cell $c_\alpha \in \Pi_\alpha$ per block, subject to coherence, σ-additivity, and non-principality.

Question 3.6 (σ-point-selection, bare form)**.** Given a set Ω and a family $\{\Pi_\alpha\}$ of countable partitions of Ω (the blocks of a concrete σ-orthoposet, pairwise overlapping in their common refinements), does there exist a selection assigning to each Π_α one cell $c_\alpha \in \Pi_\alpha$ such that

(a) (coherence) the selections agree on events common to several blocks;

(b) (σ-additivity) the induced $\{0,1\}$-valuation commutes with countable disjoint unions;

(c) (non-principality) there is no $\omega_0 \in \Omega$ with $c_\alpha =$ the cell of Π_α containing ω_0 for all α;

(d) (prescription) the selection realises a prescribed finite pattern that no single point realises?

A witness exists iff, on some admissible carrier, some prescribed pattern that is realisable by a merely finitely additive selection admits *no* selection satisfying (a)–(d) and no principal one either: the witness question is the localised *negative* instance of the selection problem, with finite-additive realisability retained. Two features of the bare form, obscured in lattice language, fix its position: its distance from a measurable cardinal, and its place within contextuality.

Remark 3.7 (The Boolean shadow is a measurable cardinal; the question is not)**.** For a compatible block family (a single Boolean base $\mathcal{B}$ on Ω), a coherent non-principal σ-additive selection $s\colon \mathcal{B} \to \{0,1\}$ has $U_s := s^{-1}(1)$ a non-principal σ-complete ultrafilter on $\mathcal{B}$, so $\exists\, s \implies |\Omega|$ measurable [13]. Question 3.6 imposes coherence (a) across incompatible, mutually non-distributive blocks: strictly stronger than an ultrafilter on any single block. The Boolean shadow reduces to a measurable cardinal; the non-distributive coherence is exactly what the reduction cannot cross (§4). The witness of §5 shows how far the shadow misleads: there the carrier's rigidity is supplied by an Ulam matrix on ω_1 — the classical device for proving ω_1 *non*-measurable — and the coherence is carried by a mere ultrafilter on $\mathcal{P}(\omega_1)/\mathrm{ctble}$, never asked to be σ-complete.

Remark 3.8 (Relation to sheaf-theoretic contextuality)**.** Question 3.6 concerns locally coherent $\{0,1\}$-valuations with no global section, contextuality in the sense of Abramsky–Brandenburger [2], σ-additive form [1]. The framing is standard; what was open is the conjunction the literature does not treat together: countable additivity + a non-Hilbert (abstractly concrete) carrier + a $\{0,1\}$-valued selection. Theorem 5.18 settles that conjunction affirmatively.

4 The boundary map

We locate the core between an upper boundary where it is known empty and the lower routes that fail to settle it. With the witness in hand (§5), this section reads as an explanation of the witness's shape: each boundary is a constraint the construction is forced to obey, and does.

The upper boundary is the Polish-representable case. Derr and Williamson [6] prove a σ-extension criterion for coherent probabilities on pre-Dynkin systems: when the carrier is Polish-representable (that is, Ω Polish, the generated σ-class equal to the Borel σ-algebra, with inner-regular blocks), finite coherence implies σ-extendability (their Theorem D.6, via Maharam [16]). Specialised to two-valued states, this says that on a Polish-representable carrier every finitely coherent s_0 globalises: no σ-essential contextual state exists. A witness must therefore live on a carrier that is not Polish-representable — as the one below, with $|\Omega| = \aleph_1$ and rigidity incompatible with inner regularity, does (Proposition 5.19).

A Hilbert-space analogue exists, but does not transfer down to the concrete case. On the projection lattice of $B(\ell^2(\kappa))$, Blecher–Weaver [5] show a singular countably additive pure state exists iff κ is Ulam measurable, the nearest positive precedent that countable additivity on a non-distributive σ-complete OML can carry set-theoretic strength. It does not settle the question here; three obstructions point the same way.

(1) *Wrong carrier.* For $\dim H \geq 3$ the projection lattice of $B(H)$ admits no two-valued state (Kochen–Specker [15]), so it is not concrete; the Blecher–Weaver dichotomy concerns pure states, not two-valued ones, which are absent there by Gleason [10].

(2) *Masa transfer fails.* Carrying a state-existence result to an abelian subalgebra means factoring through a masa; for the pure two-valued case this fails: Akemann–Weaver [3] give a pure state on $B(H)$ multiplicative on no masa, and Blecher–Weaver use Marcus–Spielman–Srivastava paving for exactly this reason.

(3) *The consumed object is structurally absent.* Blecher–Weaver extract their cardinal from a genuine ultrafilter on the diagonal masa $\ell^\infty(\kappa)$ (projections = subsets, intersection-closed). On a concrete non-Boolean σ-orthoposet, $\mathcal{F}_s := \{A \in \mathcal{L} : s(A) = 1\}$ is an ultrafilter for *no* two-valued state s, not even a filter: meet-closure fails already for δ_ω, since in any concrete MO_2 the atoms a, b have $a \cap b \neq \varnothing$ (else orthogonal) yet $a \wedge b = 0$, so $\delta_\omega(a) = \delta_\omega(b) = 1$, $\delta_\omega(a \wedge b) = 0$. The countably-complete ultrafilter an Ulam-style argument would consume is absent from the carrier: the same non-distributivity (lattice meet $\neq$ set intersection), one level down.

Remark 4.1 (Strength of the claim)**.** Remark 3.7 and (1)–(3) are cited facts assembled, not a non-reducibility theorem in a fixed reduction calculus. The claim is only that the two obvious bridges, "measurable cardinal" and "transfer from Blecher–Weaver via a masa", do not close the question, the obstruction in each being the same non-distributivity. The witness of §5 confirms the diagnosis from the other side: it consumes neither object. Item (3) in particular becomes load-bearing there: the *absence* of meet-closure is what permits three pairwise-incompatible assignments to cohere finitely (Remark 6.2).

The remaining route from below, faithful representation, is barred by the same obstruction. Routing the core through a faithful σ-tribe representation (embedding $\mathcal{L}$ into a σ-algebra of sets, so two-valued states become point evaluations) is unavailable: a faithful tribe-of-sets representation forces distributivity, hence Booleanness (the "Floor" [20]). σ-Loomis–Sikorski holds under the Riesz Decomposition Property (σ-complete MV-algebras, RDP effect algebras [8]), but OMLs lack RDP (MO_2 witnesses), and the available image is only σ-epimorphic, not point-separating. The trivialising route is blocked, again non-distributively.

An earlier version of this paper asserted, without proof and with a mis-scoped definition, that "segregated" carriers admit no witness. The correct statement and its proof follow; the repaired notion is gluing-theoretic, not centre-theoretic (the exemplar MO_ω has trivial centre by Remark 1.5, so the old definition excluded nothing).

Definition 4.2 (horizontal σ-sum)**.** A concrete σ-orthoposet $\mathcal{L}$ is a *horizontal σ-sum* if its blocks pairwise intersect in $\{\varnothing, \Omega\}$ and no two nonzero proper elements of distinct blocks are disjoint.

Lemma 4.3 (Horizontal σ-sums do not witness)**.** *If $\mathcal{L}$ is a horizontal σ-sum then $\Phi(\mathcal{L})$ holds (notation of §6): every finitely coherent two-valued state on a finite $\perp$-closed $B \subseteq \mathcal{L}$ extends to a σ-additive two-valued state on $\mathcal{L}$. In particular MO_κ, in any concrete realisation of this shape, admits no σ-essential contextual state.*

Proof. By hypothesis every pairwise-disjoint family in $\mathcal{L}$, in particular every countable orthogonal family, lies in a single block, so a function on $\mathcal{L}$ is a σ-additive state iff its restriction to each block is one: any tuple of blockwise σ-additive two-valued states glues to a global one, no cross-block constraint existing. Let s_0 on B be finitely coherent with global finitely additive extension μ. Each block $\mathcal{B}_\gamma$ is a Boolean σ-algebra of sets; $\mu \restriction (\mathcal{B}_\gamma \cap \langle B \rangle)$ concentrates, as in Proposition 2.1, on a nonempty atom of the finite Boolean algebra generated by $B \cap \mathcal{B}_\gamma$ inside $\mathcal{B}_\gamma$, and any Dirac of that block at a point of that atom is a blockwise σ-additive state agreeing with s_0 on $B \cap \mathcal{B}_\gamma$. Gluing the blockwise choices yields $\mu' \in \mathrm{St}_\sigma(\mathcal{L})$ with $\mu' \restriction B = s_0$ (elements of B lie in blocks, and on shared elements — only $\varnothing, \Omega$ — all choices agree). $\square$

Remark 4.4 (Segregated carriers, repaired)**.** Call a carrier *segregated* when it is a horizontal σ-sum, or more generally when every countable orthogonal family is confined to a single block whose σ-additive two-valued states rescue every finite trace as above. Lemma 4.3 shows segregated carriers satisfy Φ: loose gluing lets blockwise rescuers assemble unimpeded. A witness must be non-segregated — but the moral is sharper than "the blocks must be tightly glued." In the witness below the blocks *are* loosely glued at the level of the contextual pattern (no cross-core orthogonality exists at all, Lemma 5.6(IV)); what fails is the *input* to the gluing: one block is a Boolean σ-algebra on ω_1 containing an Ulam matrix, which admits no non-Dirac σ-additive two-valued state to contribute (Theorem 5.13). Segregation kills a witness only when blockwise rescuers exist; destroying them blockwise is precisely the remaining freedom.

5 The witness: the product Ulam carrier

Lineage. The carrier shape — ω_1 times a countable fibre, coupled through the countable ideal, with Ulam–Solovay rigidity making every σ-additive state countably carried — goes back to

Binder and Navara [4] (Theorem 4; restated as [18, Thm. 17.4]), who drew from it the opposite moral: such enlargements are "almost Boolean" (lattice state space, Jauch–Piron). Their framework is σ-additive throughout; the finitely additive two-valued layer — the coherent Specker pattern of §5.4, built from an ultrafilter through the parity code — does not appear there, and it is exactly this layer that turns the carrier shape into a witness. The empty-kernel triple itself descends from the Navara–Pták example [19] (in the lineage of Dravecký–Šipoš [7]), where it is used in the opposite direction: there a σ-additive non-concentrated rescuer exists; here rigidity provably destroys every one.

Throughout, M denotes ω_1 (a set of cardinality $\aleph_1$ with a fixed well-order of type ω_1), $F = \{1, 2, 3, 4\}$ is a four-element *fibre* set, and $\Omega = M \times F$. For $E \subseteq \Omega$ and $f \in F$ the *fibre trace* is $E_f = \{x \in M : (x, f) \in E\}$. For $X, Y \subseteq M$ write $X \approx Y$ when $X \,\Delta\, Y$ is countable, and $X^{(0)} = X$, $X^{(1)} = M \setminus X$. Let

$$E_4 \;=\; \{0000,\ 1111\} \cup \{\text{the six weight-2 vectors}\} \;\subseteq\; \mathbb{F}_2^4$$

be the even-weight code, and $N = \{\kappa \in E_4 : \kappa_4 = 0\} = \{0000, 1100, 1010, 0110\}$ the *normalised* patterns, one from each complement pair $\{\kappa, \kappa \,\Delta\, 1111\}$. The *cores* are

$$A_1 = M \times \{1, 2\}, \qquad A_2 = M \times \{1, 3\}, \qquad A_3 = M \times \{2, 3\},$$

with trace patterns 1100, 1010, 0110 in the coordinate order $(1, 2, 3, 4)$. Then $A_1 \cap A_2 = M \times \{1\}$, $A_1 \cap A_3 = M \times \{2\}$, $A_2 \cap A_3 = M \times \{3\}$, $A_1 \cap A_2 \cap A_3 = \varnothing$, and $(A_1 \cup A_2 \cup A_3)^{\perp} = M \times \{4\}$: the four fibres are exactly the four inhabited Boolean regions of the triple, the three "solo" regions being empty. This is the incidence geometry of Specker's parable and of Example 3.4, realised with every inhabited region uncountable.

5.1 The carrier

Lemma 5.1 (Ulam matrix). *There is a family $\{C_{\alpha,n} : \alpha < \omega_1,\ n < \omega\}$ of subsets of M such that:*

(1) *(row disjointness) $C_{\alpha,n} \cap C_{\alpha,m} = \varnothing$ for $n \neq m$;*

(2) *(row covering) $M \setminus \bigcup_{n<\omega} C_{\alpha,n} = \{\beta : \beta \leq \alpha\}$, a countable set;*

(3) *(column disjointness, exact) $C_{\alpha,n} \cap C_{\beta,n} = \varnothing$ for $\alpha \neq \beta$.*

Proof. For each $\beta < \omega_1$ fix an injection $g_\beta \colon \{\gamma : \gamma < \beta\} \to \omega$ (initial segments are countable; simultaneous choice by AC), and put $C_{\alpha,n} = \{\beta : \alpha < \beta < \omega_1,\ g_\beta(\alpha) = n\}$ [22]. (1): $g_\beta(\alpha)$ is a single value. (2): every $\beta > \alpha$ lies in $C_{\alpha, g_\beta(\alpha)}$. (3): $\beta \in C_{\alpha,n} \cap C_{\alpha',n}$ with $\alpha \neq \alpha'$ would give $g_\beta(\alpha) = n = g_\beta(\alpha')$, contradicting injectivity. □

Definition 5.2 (the carrier). Let the *cells* be $D_{\alpha,n} := C_{\alpha,n} \times F \subseteq \Omega$, and let $\mathcal{L}$ be the least σ-class on Ω containing every singleton $\{\omega\}$, every cell $D_{\alpha,n}$, and the cores A_1, A_2, A_3.

Rows ignore the fibre coordinate entirely; cores ignore the M coordinate entirely. This separation is the design principle: the Ulam machinery and the contextual geometry are prevented from interacting except through countable sets, and Theorem 5.8 is the proof that the σ-class operations cannot manufacture an interaction.

Lemma 5.3 (σ-class basics). *In any σ-class $\mathcal{L}$ containing all singletons:*

(1) *(relative complement) $X \subseteq Y$, $X, Y \in \mathcal{L} \implies Y \setminus X \in \mathcal{L}$;*

(2) *every countable $N \subseteq \Omega$ is in $\mathcal{L}$;*

(3) *(countable perturbation) $X \in \mathcal{L}$, N countable $\implies X \cup N,\ X \setminus N \in \mathcal{L}$;*

(4) *every two-valued state on $\mathcal{L}$ is monotone;*

(5) *every countable set is compatible with every element of $\mathcal{L}$.*

Proof. (1) X and $Y^\perp$ are disjoint, so $X \cup Y^\perp \in \mathcal{L}$; its complement is $Y \setminus X$. (2) A countable disjoint union of singletons. (3) $X \cup N = X \sqcup (N \setminus X)$; and $X \setminus N = (X^\perp \cup N)^\perp$ with $X^\perp \cup N \in \mathcal{L}$ by the previous sentence. (4) By (1), $s(Y) = s(X) + s(Y \setminus X) \geq s(X)$ for $X \subseteq Y$. (5) For countable N and $X \in \mathcal{L}$: $X = (X \setminus N) \sqcup (X \cap N)$ and $N = (N \setminus X) \sqcup (X \cap N)$ with all parts in $\mathcal{L}$ by (2)–(3), a joint orthogonal decomposition. □

5.2 The trace invariant

Definition 5.4 (the invariant). Say $E \subseteq \Omega$ *satisfies the invariant*, written $E \in \widetilde{P}$, if there exist $\xi \subseteq M$ and $\kappa \in E_4$ with

$$E_f \;\approx\; \xi^{(\kappa_f)} \qquad (f = 1, 2, 3, 4);$$

call (ξ, κ) a *representation* of E. Intrinsically: $E \in \widetilde{P}$ iff all pairwise symmetric differences $E_f \,\Delta\, E_g$ are countable or co-countable and the induced parity vector is even.

Lemma 5.5 (representation rigidity). *If (ξ, κ) and (η, λ) both represent E with $\kappa, \lambda \in E_4$, then either $\lambda = \kappa$ and $\eta \approx \xi$, or $\lambda = \kappa \,\Delta\, 1111$ and $\eta \approx \xi^\perp$. Consequently:*

(1) *each $E \in \widetilde{P}$ has a unique representation with $\kappa \in N$, up to $\approx$-change of ξ;*

(2) *E is countable iff its normalised representation is $(\xi, 0000)$ with $\xi \approx \varnothing$;*

(3) *if $\kappa \in N \setminus \{0000\}$ (a* weight-2 *element) then E is uncountable: some trace is $\approx \xi$ and some $\approx \xi^\perp$, and these cannot both be countable.*

Proof. If $\kappa_f = \lambda_f$ for some f, then $\xi^{(\kappa_f)} \approx \eta^{(\kappa_f)}$ gives $\xi \approx \eta$; substituting into the remaining coordinates forces $\kappa_g = \lambda_g$ for all g, since $\xi \approx \xi^\perp$ is impossible, M being uncountable. Otherwise $\kappa_f \neq \lambda_f$ for all f, i.e. $\lambda = \kappa \,\Delta\, 1111$, and any coordinate gives $\eta \approx \xi^\perp$. Exactly one of $\kappa, \kappa \,\Delta\, 1111$ has fourth coordinate 0, giving (1); (2), (3) are immediate. □

Lemma 5.6 (disjointness table). *Let $E, G \in \widetilde{P}$ have normalised representations (ξ, κ), (η, λ), $\kappa, \lambda \in N$, and suppose $E \cap G = \varnothing$ (only $E_f \cap G_f$ countable, for each f, is used). Coordinatewise, disjointness mod countable of $\xi^{(\kappa_f)}$ and $\eta^{(\lambda_f)}$ reads*

$(\kappa_f, \lambda_f) = (0,0)$: $\xi \cap \eta \approx \varnothing$; $(0,1)$: $\xi \subseteq \eta$ *mod ctble*; $(1,0)$: $\eta \subseteq \xi$ *mod ctble*; $(1,1)$: $\xi \cup \eta \approx M$.

The ten unordered pairs (κ, λ) resolve as:

Case	(κ, λ)	*Forced relations*	*Conclusion*
(I)	$(0000, 0000)$	$\xi \cap \eta \approx \varnothing$	$E \sqcup G \in \widetilde{P}$, *rep.* $(\xi \cup \eta,\ 0000)$
(II)	(κ, κ), κ *weight-2*	$\xi \cap \eta \approx \varnothing$ and $\xi \cup \eta \approx M$	$\eta \approx \xi^\perp$; $E \sqcup G$ *co-ctble, rep.* $(M, 0000)$
(III)	$(0000, \lambda)$, λ *weight-2*	$\xi \subseteq \eta$, $\xi \cap \eta \approx \varnothing$ *mod ctble*	$\xi \approx \varnothing$: E countable; *rep. of* $E \sqcup G$ *is* (η, λ)
(IV)	$\kappa \neq \lambda$, *both weight-2*	$\xi \approx \eta$, $\xi \cap \eta \approx \varnothing$, $\xi \cup \eta \approx M$	impossible *($\varnothing \approx M$)*

Proof. The coordinate translations are immediate; (I)–(III) are read off. For (IV): any two distinct elements of $N \setminus \{0000\}$ agree in exactly one of the first three coordinates and both vanish in the fourth, so the multiset of coordinate pairs (κ_f, λ_f) is exactly $\{(1,1), (1,0), (0,1), (0,0)\}$. Then $(1,0)$ and $(0,1)$ give $\xi \approx \eta$; $(0,0)$ gives $\xi \cap \eta \approx \varnothing$, hence $\xi \approx \varnothing$; $(1,1)$ gives $\xi \cup \eta \approx M$, hence $\varnothing \approx M$ — false, M being uncountable. □

Row (IV) is the theorem in miniature: no element of $\mathcal{L}$ carrying one core pattern is ever exactly disjoint from an element carrying a different core pattern, beyond countable degeneracy. This is what makes the three simultaneous assignments $s_0(A_i) = 1$ immune to additivity constraints; the parity-triangle certificate that dooms tighter designs (three exact events with $a + b = a + c = b + c = 1$ over $\{0, 1\}$, summing to the odd total 3) cannot form.

Lemma 5.7 (family trichotomy)**.** *Let $\{E_i\}_{i<\omega} \subseteq \widetilde{P}$ be pairwise exactly disjoint. Then exactly one of:*

1. *every E_i of uncountable class is diagonal ($\kappa_i = 0000$, $\xi_i \not\approx \varnothing$), pairwise $\xi_i \cap \xi_j \approx \varnothing$;*
2. *exactly one E_{i_0} is weight-2; all other E_i are countable;*
3. *exactly two E_{i_0}, E_{i_1} are weight-2, in the same coset with $\xi_{i_1} \approx {\xi_{i_0}}^{\perp}$; all other E_i are countable.*

Proof. By Lemma 5.6(IV) all weight-2 members share one coset. By (II) two same-coset weight-2 members are complementary mod countable; a third in that coset would be $\approx$-complementary to both, giving $\xi \approx \xi^{\perp}$ — impossible. So there are at most two weight-2 members. If at least one exists, any diagonal member is disjoint from it, hence countable by (III); if two (complementary) weight-2 members exist, any further member has each trace class below the complement of their join, which is $\approx \varnothing$, hence is countable. □

Theorem 5.8 (normal form)**.** $\mathcal{L} \subseteq \widetilde{P}$*: every element of the carrier satisfies the invariant.*

Proof. $\widetilde{P}$ contains the generators: singletons and $\varnothing$ are $(\varnothing, 0000)$; Ω is $(M, 0000)$; cells are $(C_{\alpha,n}, 0000)$; the cores are $(\varnothing, 1100)$, $(\varnothing, 1010)$, $(\varnothing, 0110)$. It is complement-closed: $(E^{\perp})_f = (E_f)^{\perp}$, so $(\xi, \kappa \,\Delta\, 1111)$ represents $E^{\perp}$. It is closed under countable pairwise exactly-disjoint unions, by Lemma 5.7: in case (1) the union is represented by $(\bigcup_i \xi_i^*, 0000)$, where $\xi_i^* := \xi_i \setminus \bigcup_{j<i}(\xi_i \cap \xi_j)$ trims the pairwise-countable overlaps — each $\bigcup_{j<i}(\xi_i \cap \xi_j)$ is countable, so $\xi_i^* \approx \xi_i$, the ξ_i^* are exactly disjoint, and coordinatewise the union's trace differs from $\bigcup_i \xi_i^*$ by a countable union of countable sets; in case (2) by $(\xi_{i_0}, \kappa_{i_0})$, the countable members perturbing each trace countably; in case (3) by $(M, 0000)$. Hence $\{E : E \in \widetilde{P}\}$ is a σ-class containing the generators, so it contains $\mathcal{L}$. □

Corollary 5.9 (the missing meet region)**.** *No element of $\mathcal{L}$ is $\approx$-equal (fibrewise) to $A_1 \cap A_2 = M \times \{1\}$; in particular $A_1 \cap A_2 \notin \mathcal{L}$.*

Proof. The trace classes of $M \times \{1\}$ are $(M, \varnothing, \varnothing, \varnothing)$ mod countable. In any representation, the coordinates take at most the two values $\xi, \xi^{\perp}$ mod countable; here they take the classes of M and $\varnothing$, so $\xi \approx \varnothing$ or $\xi \approx M$, and the pattern is κ or $\kappa \,\Delta\, 1111$ for some $\kappa \in E_4$. But 1000 and 0111 have odd weight. □

Corollary 5.10 (stripping bar)**.** *If $E \in \mathcal{L}$ has E_3, E_4 countable (i.e. $E \subseteq A_1$ mod countable), then E is countable or $E \approx A_1$ fibrewise. Likewise for A_2, A_3 and the core complements. If $E \subseteq A_1 \cap A_2$ exactly, E is countable.*

Proof. Take the normalised (ξ, κ); throughout, coordinate f carries $\xi^{(\kappa_f)}$ (Definition 5.4). If $\kappa = 0000$: coordinates $3, 4$ carry ξ, so $\xi \approx \varnothing$ and E is countable. If $\kappa = 1100$: coordinates $3, 4$ carry ξ ($\kappa_3 = \kappa_4 = 0$), so $\xi \approx \varnothing$; coordinates $1, 2$ then carry $\xi^{\perp} \approx M$, so $E \approx A_1$. If $\kappa = 1010$ or 0110: one of coordinates $3, 4$ carries ξ and the other $\xi^{\perp}$, forcing $\xi \approx \varnothing$ and $\xi \approx M$ simultaneously — impossible. For the last claim, $E \subseteq M \times \{1\}$ makes coordinates $2, 3, 4$ countable, and the same analysis kills all three weight-2 cosets, leaving $\xi \approx \varnothing$. □

Corollary 5.11 (incompatibility; not a lattice). *A_1 and A_2 are incompatible in $\mathcal{L}$; hence $\mathcal{L}$ is non-Boolean. Moreover $A_1 \wedge A_2$ does not exist: the lower bounds of $\{A_1, A_2\}$ in $\mathcal{L}$ are exactly the countable subsets of $M \times \{1\}$, among which there is no maximum. Thus $\mathcal{L}$ is a σ-complete orthomodular poset and provably not a lattice.*

Proof. Compatibility in a concrete orthomodular poset means $A_1 = A' \sqcup D$, $A_2 = A'' \sqcup D$ with $A', A'', D \in \mathcal{L}$ pairwise disjoint; then $D = A_1 \cap A_2$ (expand $A_1 \cap A_2$ using $A' \cap A'' = \varnothing$), contradicting Corollary 5.9. The lower-bound computation is the last clause of Corollary 5.10. □

Corollary 5.12 (centre). *$Z(\mathcal{L}) = \{E \in \mathcal{L} : E$ countable or co-countable$\}$: the quotient of $\mathcal{L}$ by the countable ideal has trivial centre, so $\mathcal{L}$ is essentially irreducible.*

Proof. ($\supseteq$) Lemma 5.3(5) and complement-closure of the centre. ($\subseteq$) Let E be central. Compatibility with A_1 puts $E \cap A_1 \in \mathcal{L}$ (as in Corollary 5.11, the common part of a compatible decomposition is the intersection), whose trace quadruple is $(E_1, E_2, \varnothing, \varnothing)$ with the last two coordinates exactly empty. Membership in $\widetilde{P}$ (Theorem 5.8) then forces, by the analysis of Corollary 5.10, $[E_1] = [E_2] \in \{0, 1\}$ as classes mod countable. Compatibility with A_2 gives $[E_1] = [E_3]$, with A_3 gives $[E_2] = [E_3]$, all in $\{0, 1\}$; so the first three trace classes are a common constant t. Membership of E itself in $\widetilde{P}$ forces the fourth class to equal t: with three coordinates constant t, the pairwise-difference and parity conditions of Definition 5.4 require the fourth difference to vanish. Thus E is countable ($t = 0$) or co-countable ($t = 1$).

For the quotient claim, let $[E]$ be central in $\mathcal{L}$/ctble. Compatibility of $[E]$ with $[A_1]$ in the quotient supplies $D \in \mathcal{L}$ with $D \approx E \cap A_1$ (the quotient decomposition, as in Corollary 5.11, computed mod countable). Then $D \in \widetilde{P}$ has coordinates $3, 4$ countable, and the analysis of Corollary 5.10 — which only ever uses trace *classes* — runs unchanged mod countable, giving $[E_1] = [E_2] \in \{0, 1\}$. Compatibility with $[A_2]$, $[A_3]$ give $[E_1] = [E_3]$ and $[E_2] = [E_3]$, and membership of E in $\widetilde{P}$ forces the fourth class exactly as above. Hence E is countable or co-countable, i.e. $[E] \in \{0, 1\}$: the quotient centre is trivial. □

5.3 Rigidity

Theorem 5.13 (rigidity). *Every σ-additive two-valued state on $\mathcal{L}$ is Dirac.*

Proof. Let s be σ-additive and two-valued. If $s(\{\omega\}) = 1$ for some ω, monotonicity (Lemma 5.3(4)) gives $s(E) = 1$ whenever $\omega \in E$, and $\omega \notin E$ gives $s(E^\perp) = 1$, so $s(E) = 0$: $s = \delta_\omega$. Otherwise s vanishes on all singletons, hence, by σ-additivity, on all countable sets. Fix a row α: the cells $\{D_{\alpha,n}\}_{n<\omega}$ are pairwise exactly disjoint (Lemma 5.1(1)), their union U_α lies in $\mathcal{L}$ (σ-class closure), and $\Omega \setminus U_\alpha$ is countable (Lemma 5.1(2)), so $s(U_\alpha) = 1$ and σ-additivity over the row yields exactly one $n(\alpha) < \omega$ with $s(D_{\alpha,n(\alpha)}) = 1$. The map $\alpha \mapsto n(\alpha)$ sends ω_1 into ω, so some value n^* has uncountable fibre; fix $\alpha \neq \beta$ with $n(\alpha) = n(\beta) = n^*$. By column disjointness (Lemma 5.1(3)), $D_{\alpha,n^*} \cap D_{\beta,n^*} = \varnothing$ exactly; their union is in $\mathcal{L}$, and additivity gives s the value 2 there — contradiction. So the second case is void. □

The proof consumes only singletons, cells, rows, columns, and σ-class closure — nothing about cores or the invariant. Rigidity is therefore robust to any enlargement of the generator set preserving these items. Note the inversion of Remark 3.7: the Ulam matrix, classically the device by which ω_1 fails to be measurable, here works *for* the witness — clause (ii) of Theorem 3.2 is achieved outright, in ZFC, by importing the matrix into the carrier as generators.

5.4 The state

Definition 5.14. Fix an ultrafilter $\mathcal{U}$ on M extending the co-countable filter (the filter is proper, M being uncountable), and write $u(\xi) \in \{0, 1\}$ for its characteristic function. Then u

vanishes on countable sets, is constant on $\approx$-classes, satisfies $u(\xi^\perp) = 1 - u(\xi)$, and is additive on pairs with $\xi \cap \eta \approx \varnothing$. For $E \in \mathcal{L}$ with normalised representation (ξ, κ) define the *vote*

$$v(E) \;:=\; \kappa\,\Delta\,\big(u(\xi)\cdot 1111\big) \;\in\; E_4,$$

and, with $\hat{S} := \{1111,\ 1100,\ 1010,\ 0110\}$ (one member from each complement pair of E_4),

$$m(E) \;:=\; \mathbf{1}\big[\,v(E) \in \hat{S}\,\big].$$

Lemma 5.15 (well-definedness)**.** *v, hence m, is independent of the representation. Moreover $v(E^\perp) = v(E)\,\Delta\,1111$, so $m(E^\perp) = 1 - m(E)$; $m(\Omega) = 1$; m vanishes on countable sets; and $m(A_1) = m(A_2) = m(A_3) = 1$.*

Proof. By Lemma 5.5 the normalised representation is unique up to $\approx$-change of ξ, on which u is constant; computing from the non-normalised partner $(\xi^\perp, \kappa\,\Delta\,1111)$ gives $\kappa\,\Delta\,1111\,\Delta\,(u(\xi^\perp)\cdot 1111) = \kappa\,\Delta\,((1+u(\xi^\perp))\cdot 1111) = \kappa\,\Delta\,(u(\xi)\cdot 1111)$. Complement: representation $(\xi, \kappa\,\Delta\,1111)$, vote $v(E)\,\Delta\,1111$; since $\hat{S}$ selects one member per complement pair, m flips. Ω: $(M, 0000)$, $u(M) = 1$, vote $1111 \in \hat{S}$. Countable E: $(\varnothing, 0000)$, vote $0000 \notin \hat{S}$. Cores: $\xi = \varnothing$, $u = 0$, votes $1100, 1010, 0110 \in \hat{S}$. $\square$

Theorem 5.16 (m is a two-valued state)**.** *For every finite pairwise-disjoint family $\{E_i\} \subseteq \mathcal{L}$ (whose union is automatically in $\mathcal{L}$), $m(\bigsqcup_i E_i) = \sum_i m(E_i)$. Hence m is a finitely additive two-valued state on $\mathcal{L}$ extending the pattern $s_0(A_i) = 1$, which is therefore finitely coherent on $\mathcal{L}$.*

Proof. Pairs suffice (finite families follow by induction, partial unions lying in $\mathcal{L}$). Let $E \cap G = \varnothing$, $H = E \sqcup G$, and run Lemma 5.6 on normalised representations. (I): $m(E) = u(\xi)$ and $m(G) = u(\eta)$ (votes $u \cdot 1111$; $0000 \notin \hat{S}$, $1111 \in \hat{S}$), and H is represented by $(\xi \cup \eta, 0000)$ up to countable trimming, so $m(H) = u(\xi \cup \eta) = u(\xi) + u(\eta)$ by additivity of u on $\approx$-disjoint pairs. (II): $\eta \approx \xi^\perp$ gives $v(G) = v(E)\,\Delta\,1111$, so exactly one of $m(E), m(G)$ is 1; and $H \approx \Omega$ has vote 1111, $m(H) = 1$. (III): the countable member has $m = 0$ and H shares the uncountable member's normalised representation (countable perturbation), so $m(H) = m(G) + 0$. (IV) cannot occur. Finally, s_0 is a two-valued state on the $\perp$-closure $B = \{\varnothing, \Omega, A_i, A_i{}^\perp\}$ independently of m: each pairwise intersection among $\{A_i, A_i{}^\perp\}_{i \le 3}$ $(i \ne j)$ is a full fibre $M \times \{f\}$, nonempty, so the only additivity constraints inside B are the complement pairs; and $m \restriction B = s_0$ by Lemma 5.15. $\square$

Remark 5.17 (the σ-failure of m is self-witnessing)**.** Suppose u selected a cell in every row: $\forall \alpha\, \exists n(\alpha)$ with $u(C_{\alpha, n(\alpha)}) = 1$. Pigeonholing $\alpha \mapsto n(\alpha)$ as in Theorem 5.13 produces two disjoint sets of u-value 1 — impossible for an ultrafilter. So some row α_0 has $u(C_{\alpha_0, n}) = 0$ for all n: there $m(D_{\alpha_0, n}) = 0$ for every n while m of the (co-countable) row union is 1. Every ultrafilter fails σ-additivity on some row, by the same pigeonhole that drives rigidity; no genericity demands on the matrix are needed anywhere in the construction.

5.5 The main theorem

Theorem 5.18 (ZFC witness)**.** *Let $\mathcal{L}$, B, s_0 be as above. Then:*

(1) *s_0 is finitely coherent on $\mathcal{L}$: m extends it (Theorem 5.16);*

(2) *no σ-additive two-valued state on $\mathcal{L}$ extends s_0: by Theorem 5.13 any such extension is some δ_ω, and by Lemma 3.1 this requires $\omega \in K(s_0) = A_1 \cap A_2 \cap A_3 = \varnothing$;*

(3) *hence s_0 is a σ-essential contextual state on $\mathcal{L}$* (Definition 1.2)*, and the target sentence Ψ of §6 is a theorem of ZFC.*

Proposition 5.19 (admissibility)**.** *The carrier $\mathcal{L}$ is concrete and σ-complete (by construction); non-Boolean, with the incompatibility of A_1, A_2 robust under countable perturbation (Corollaries 5.9, 5.11); essentially irreducible, with centre exactly the countable/co-countable sets (Corollary 5.12), which m annihilates; non-segregated in the sense of Remark 4.4, since the diagonal block — the Boolean σ-algebra on ω_1 generated by singletons and cells — admits no non-Dirac σ-additive two-valued state to contribute to any gluing; and not Polish-representable, as a corollary of the witness property via Derr–Williamson (§4). It is not a lattice (Corollary 5.11).*

6 Discussion

To state what has been proved in the notation of the earlier version: write $\mathrm{St}(B)$ for the two-valued states on a sub-orthoposet B, $\mathrm{St}_\sigma(\mathcal{L})$ for the global σ-additive two-valued states on $\mathcal{L}$, $\mathrm{St}_{fa}(\mathcal{L})$ for the global finitely additive ones, $\mathrm{Fin}_\perp(\mathcal{L})$ for the finite $\perp$-closed sub-orthoposets, and Adm for the admissible carriers: concrete, σ-complete, non-Boolean, essentially irreducible, non-segregated, non-Polish-representable (the last two being consequences of witnesshood, by Lemma 4.3 and Derr–Williamson, not extra demands). The *σ-additive lifting* property and its negation are

$$\Phi(\mathcal{L}) :\equiv \ \forall B \in \mathrm{Fin}_\perp(\mathcal{L})\ \forall s \in \mathrm{St}(B)\ \Big[\exists \mu \in \mathrm{St}_{fa}(\mathcal{L})\ \mu\restriction B = s \implies \exists \nu \in \mathrm{St}_\sigma(\mathcal{L})\ \nu\restriction B = s\Big],$$

$$\Psi \ :\equiv\ \exists \mathcal{L} \in \mathrm{Adm}\ \neg\Phi(\mathcal{L}).$$

Theorem 5.18 proves Ψ in ZFC. Proposition 2.1 is the statement that Boolean carriers satisfy Φ; Lemma 4.3 that horizontal σ-sums do; Derr–Williamson that Polish-representable carriers do. The witness therefore inhabits precisely the cell the boundary map of §4 isolated, and each constraint of that map is visible in its anatomy.

Remark 6.1 (consistency strength, resolved)**.** An earlier version of this paper argued that the strength of Ψ was open in both directions — the measurable-cardinal lower bound failing because the reduce-to-Dirac argument is a *σ-algebra* theorem unavailable on σ-classes, the Blecher–Weaver upper bound failing to transfer for want of a masa. Theorem 5.18 settles this in the most deflationary way: Ψ is a theorem of ZFC and carries no large-cardinal strength whatsoever. The earlier analysis survives as the explanation. On the lower-bound side, the reduce-to-Dirac argument is not escaped but *embraced*: the carrier imports exactly the fragment of the σ-algebra needed to run Ulam (rows, columns, singletons), and Theorem 5.8 certifies the fragment does not Booleanise the carrier. On the upper-bound side, the object Blecher–Weaver consume — a σ-complete ultrafilter — is never needed: coherence is carried by an arbitrary ultrafilter on $\mathcal{P}(\omega_1)/\mathrm{ctble}$, a BPI-level object, of which σ-additivity is never demanded (Remark 5.17 shows it necessarily fails).

Remark 6.2 (where the coherence lives)**.** Section 4(3) observed that on a concrete non-Boolean carrier $\mathcal{F}_s$ is never a filter — the countably-complete ultrafilter an Ulam-style transfer would consume is structurally absent. The witness shows the carrier does not need to *host* that object: it borrows only the finite trace of an ultrafilter one Boolean level down, through the vote map of Definition 5.14. The same non-distributivity that blocks the upper bound (no meet-closure) is what lets the three core assignments coexist: Lemma 5.6(IV) says the additivity constraints that would link them do not exist in $\mathcal{L}$. Relatedly, the constraint of the earlier version's Remark on witness technology stands but must be scoped: the *incompatibility* of a witness cannot be imported from a Boolean-ambient combinatorial skeleton (in such an ambient a maximal coherent $\{0, 1\}$-assignment is an ultrafilter, hence meet-closed, and the pattern globalises); the witness's incompatibility is native, carried by the fibre parity code. What *can* be imported from the Boolean ambient is rigidity — the destruction of non-Dirac σ-states — and the construction is exactly this division of labour, with Theorem 5.8 as the proof that the imported engine and the native geometry cannot interact.

Remark 6.3 (a quotient with no σ-additive states at all)**.** In the quotient of $\mathcal{L}$ by the countable ideal, Dirac states die (they concentrate on singletons) and non-Dirac σ-additive states are barred by Theorem 5.13. The quotient is thus a σ-complete non-Boolean orthomodular poset carrying finitely additive two-valued states (the induced $\bar{m}$) but *no* σ-additive two-valued state whatsoever. This artifact should be graded modestly: *abstract* logics with finitely additive states that are not countably additive are established art — Navara [17] builds a σ-orthocomplete lattice logic on which regular finitely additive states fail σ-additivity (via stateless Greechie components, hence nothing set-representable), and Dvurečenskij–Neubrunn–Pulmannová [9] exhibit the phenomenon on the splitting-subspace logic of an incomplete inner-product space (not set-representable, not σ-orthocomplete). The quotient's residual interest is the conjunction it inherits from the witness: *two-valued* states, σ-completeness, and provenance as a quotient of a *concrete* carrier. It is not itself concrete ($\mathrm{St}_\sigma = \varnothing$, while concrete logics always carry Diracs).

What remains open

Question 6.4 (the lattice form)**.** Does there exist a concrete σ-complete orthomodular *lattice* carrying a σ-essential contextual state? Equivalently: does latticehood imply Φ? We conjecture it does. Three independent mechanisms destroy meets exactly where contextual structure enters — pasted-loop concrete closures, the even-cardinality logic, and the singletons-below-fibres mechanism of Corollary 5.11 — and the rigidity method of Theorem 5.13 appears to require singletons, which on this carrier are what break the meet. If the conjecture holds, the OMP/OML distinction, usually bookkeeping, is precisely the boundary of σ-essential contextuality.

Question 6.5 (cardinality)**.** The witness has $|\Omega| = \aleph_1$. Wright's theorem bars finite witnesses; can a countable Ω carry one (necessarily without all singletons, or with a different rigidity mechanism)? Is there a ZFC lower bound beyond "uncountable"?

Question 6.6 (characterising Φ)**.** Boolean carriers, Polish-representable carriers, and horizontal σ-sums satisfy Φ; the product Ulam carrier does not. Is there a structural characterisation of Φ for concrete σ-orthoposets — a gluing-theoretic or exhaustion-theoretic condition separating the two classes, with Theorem 5.18 as the canonical failure?

Acknowledgements. The author thanks the anonymous referees (forthcoming). The construction of §5 was developed jointly with a large language model (Anthropic Claude) in interactive sessions, which also assisted with literature search, verification of cited statements against primary sources, and exposition; OpenAI ChatGPT assisted with earlier literature work. All mathematical claims are being independently verified by the author, including a Lean 4 formalization in progress; see the Verification status paragraph in §1.

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